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COMPUTED TOMOGRAPHY

Ureteral calculi lithotripsy for single ureteral calculi: can DNN-assisted model help preoperatively predict risk factors for sepsis?

Mingzhen Chen[1] · Jiannan Yang[2] · Junlin Lu[3] · Ziling Zhou[4] · Kun Huang[5] · Sihan Zhang[3] · Guanjie Yuan[1] · Qingpeng Zhang[2] · Zhen Li[1]

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Abstract

Objectives To explore the utility of radiomics and deep learning model in assessing the risk factors for sepsis after flexible ureteroscopy lithotripsy (FURL) or percutaneous nephrolithotomy (PCNL) in patients with ureteral calculi.

Methods This retrospective analysis included 847 patients with treatment-naive proximal ureteral calculi who received FURL or PCNL. All participants were preoperatively conducted non-contrast computed tomography scans, and relevant clinical information was meanwhile collected. After propensity score matching, the radiomics model was established to predict the onset of sepsis. A deep learning model was also adapted to further improve the prediction accuracy. Performance of these trained models was verified in another independent external validation set including 40 cases of ureteral calculi patients.

Results The overall incidence of sepsis after FURL or PCNL was 5.9%. The least absolute shrinkage and selection operator (LASSO) regression analysis revealed 26 predictive variables, with an overall AUC of 0.881 (95% CI, 0.813–0.931) and an AUC of 0.783 (95% CI, 0.766–0.801) in external validation cohort. Judicious adaption of a deep neural network (DNN) model to our dataset improved the AUC to 0.920 (95% CI, 0.906–0.933) in the internal validation. To eliminate the overfitting, external validation was carried out for DNN model (AUC = 0.874 (95% CI, 0.858–0.891)).

Conclusions The DNN was more effective than the LASSO model in revealing risk factors for sepsis after FURL or PCNL in single ureteral calculi patients, and females are more susceptible to sepsis than males. Deep learning models have the potential to act as gatekeepers to facilitate patient stratification.

Key Points

- *Both the least absolute shrinkage and selection operator (LASSO) and deep neural network (DNN) models were shown to be effective in sepsis prediction.*
- *The DNN model achieved superior prediction capability, with an AUC of 0.920 (95% CI, 0.906–0.933).*
- *DNN-assisted model has potential to serve as a gatekeeper to facilitate patient stratification.*

Keywords Percutaneous nephrolithotomy · Flexible ureteroscopy · Ureteral calculi · Computed tomography · Lithotripsy

Abbreviations

AI	Artificial intelligence
AUC	Area under curve
BMI	Body mass index
CT	Computed tomography
DNN	Deep neural network

✉ Qingpeng Zhang
qingpeng.zhang@cityu.edu.hk

✉ Zhen Li
zhenli@hust.edu.cn

[1] Department of Radiology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, No.1095 Jiefang Avenue, Wuhan, Hubei 430030, China

[2] School of Data Science, City University of Hong Kong, Kowloon, Hong Kong 999077, China

[3] Department of Urology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

[4] Biomedical Engineering Department, College of Life Science and Technology, Huazhong University of Science and Technology, Wuhan, China

[5] School of Information and Technology, University of Science and Technology of China, Hefei, China

FURL	Flexible ureteroscopy lithotripsy
HU	Hounsfield unit
ICC	Intraclass correlation coefficient
LASSO	The least absolute shrinkage and selection operator
NCT	Non-enhanced CT
PCNL	Percutaneous nephrolithotomy
PSM	Propensity score matching
ROC	Receiver operating characteristic
ROI	Regions of interest

Introduction

Urolithiasis is the third most common diseases in the urinary system, and the prevalence is still rising [1, 2]. Currently, there are many treatment options for ureteral stones. With the continuous innovation and development of various auxiliary equipment, traditional open surgical procedures have been replaced by various types of minimally invasive surgical methods, such as flexible ureteroscopy lithotripsy (FURL) and percutaneous nephrolithotomy (PCNL) [3]. However, intracavitary operation may cause some complications, such as urosepsis. Although sepsis is not common, it is the most dangerous one. Without appropriate identification and intervention, the condition may worsen, and even lead to shock and organ dysfunction [4, 5].

Studies have shown that some clinical factors are related to the incidence of sepsis, such as positive urine culture, long stent indwelling time, and abnormal anatomical structure [4, 6]. However, there is still a lack of effective quantitative prediction models for early postoperative sepsis, which is essential to avoid its occurrence [7]. Recently, the correlation between the stone characteristics and the incidence of postoperative sepsis has received attention. According to the guidelines of the European Association of Urology [8], infectious stones are more brittle and easier to disintegrate, and various pathogenic bacteria will be released during the lithotripsy process, leading to infection or even sepsis. Studies have shown that the stone features can also aid in predicting the infection status after surgery. Eswara et al [9] found that compared with preoperative urine culture, bacterial culture of stones may be more valuable for determining sepsis, but the analysis of stone characteristics can only be done during or after the operation [10, 11].

Non-enhanced computed tomography (CT) scan is a standard examination method for ureteral stones diagnosis [12], and it provides a multi-faceted reference for patients with urinary calculi. Some researchers [13, 14] seek to analyze the stone composition based on Hounsfield unit (HU) value. However, due to partial volume effect, HU value derived from CT images cannot precisely characterize the intra-stone condition, deteriorating the reliability of this method [15]. Various studies have investigated the application of stone-specific radiomics and artificial intelligence (AI) in the precise medical management of urolithiasis, such as automatic detection, efficacy prediction, and differentiation [16–19]. Deep neural network (DNN) [20] is an artificial neural network with multiple layers between the input features and output predictions, which has already been widely used in medical research, such as diabetic retinopathy identification [21].

But so far, the research on the association between stone components and sepsis based on machine and deep learning algorithms is scarce. Herein, the purpose of our study was to determine the performance of the routine algorithm (i.e., LASSO) and DNN for the preoperative prediction of sepsis in ureteral calculi patients after undergoing FURL or PCNL.

Materials and methods

Study participants

Documents from 1,469 patients with ureteral calculi in Tongji Hospital of Huazhong University of Science and Technology who underwent PCNL or FURL between January 2012 and December 2018 were retrospectively retrieved. Criteria for selecting the subjects were as follows: (1) patients with unilateral, solitary, and proximal ureteral calculi underwent PCNL or FURL treatment; (2) age ≥ 18 years; (3) non-enhanced CT examination before operation. Exclusion criteria were as follows: (1) kidney anatomical abnormality (such as horseshoe kidney, solitary kidney, transplanted kidney, or duplicate kidney); (2) inferior CT image quality or lost to follow-up. We collected the relevant clinical data and stone characteristics of the patients, including age, gender, body mass index (BMI), comorbidities, stone size and laterality, preoperative indwelling stent, blood examination, fever, urinalysis, and urine culture.

According to the guidelines of the International Conference on the Definition of Sepsis in 2001 [22], postoperative sepsis is defined as infection occurs within 48 h after surgery, accompanied by at least the following two conditions: (I) body temperature > 38°C; (II) heart rate > 90/min; (III) respiratory rate > 20/min; (IV) white blood cell count < 4000, or > 12000 cells/μL.

A total number of 847 patients met the inclusion criteria, including 50 cases in the sepsis group and 747 cases in the non-sepsis group. In an attempt to control the influence of confounding factors (such as surgical method, age, gender, white blood cell, urine culture, and stone size), propensity score matching (PSM) was performed. Then, the matched cases were incorporated into further radiomics analysis (shown in Fig. 1). To further validate the established models, we collected 40 ureteral calculi patients treated with PCNL or FURL between January 2019 and December 2021 as an independent external validation cohort, including 10 cases in the

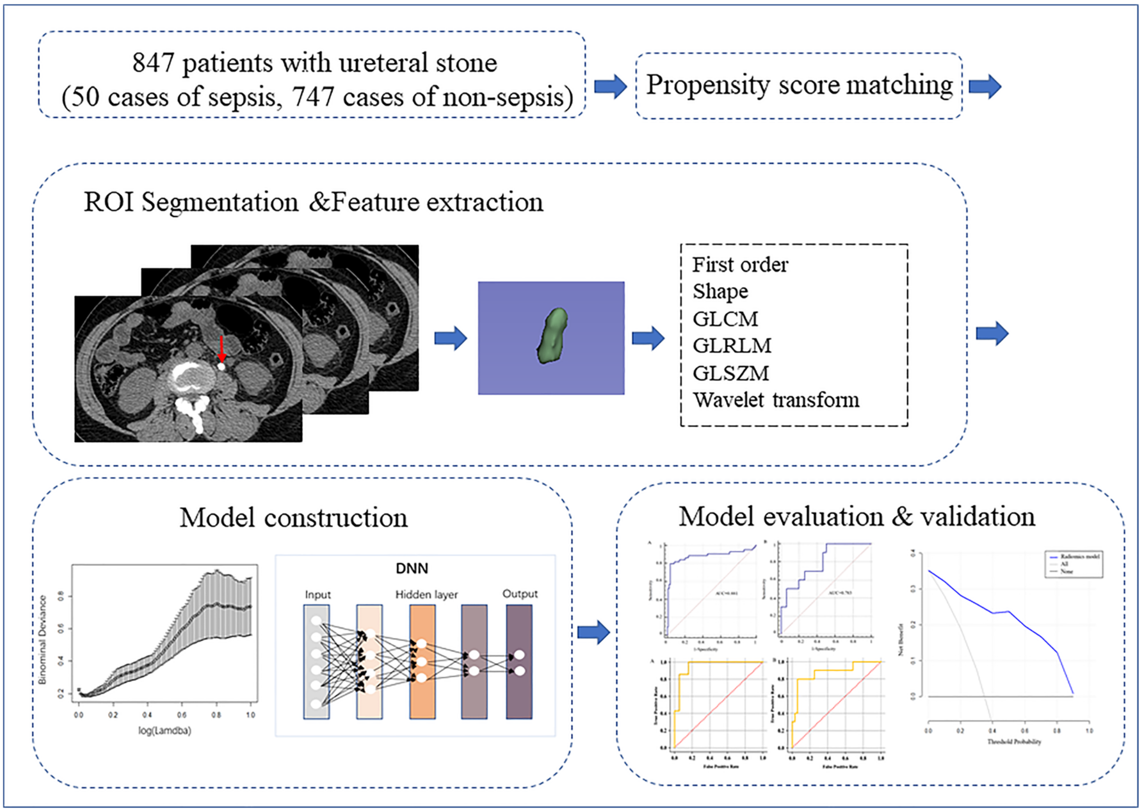

Fig. 1 Research flowchart of patients with ureteral calculi

sepsis group and 30 cases in the non-sepsis group. The inclusion and exclusion criteria of these patients were kept in line with the previous cohort.

Image acquisition and radiomics interpretation

Examinations of ureteral stones were performed on non-enhanced CT (NCCT) (Discovery CT750 HD, GE Healthcare). Pertinent CT imaging acquisition parameters are as follows: lamp voltage 100–120 kV; automatic tube current, 200–350 mA; rotation time 0.5 s; scanning slice thickness 5mm; reconstruction thickness 1.25 mm.

Using the open-source image segmentation software 3D Slicer (version 4.9.0; www.slicer.org), the regions of interest (ROIs) were manually delineated slice by slice in each axial CT image by reader 1 (Mingzhen Chen, an experienced radiologist in abdominal imaging). Meanwhile, relevant stone characteristics were recorded: (1) stone size, which is the largest diameter of the stone on the image; (2) stone laterality; and (3) hydronephrosis level. Feature extraction was carried out using internal texture analysis plug-in of 3D Slicer. To normalize the voxel spacing and control image noise, all images were resampled to $1 \times 1 \times 1$ mm^3 voxels with a fixed bin width of 25. Finally, the radiomics features included first-order texture features (mean, variance, skewness, kurtosis, etc.), three-dimensional shape and size features (sphericity, maximum three-dimensional diameter, surface area, volume, etc.), texture features (gray-level co-occurrence matrix (GLCM), neighboring gray tone difference matrix (NGTDM), etc.), and wavelet texture features (high-frequency and low-frequency information features from the decomposed image). To evaluate the reproducibility of the extracted features, reader 1 carried out the identical procedure twice on 30 randomly selected patients in a 2-week period. Reader 2 (Guanjie Yuan, a radiologist with 3 years' abdominal imaging experience) independently segmented the same 30 cases. The intraclass correlation coefficient (ICC) was used to measure the test-retest and inter-observer reliability.

Development and evaluating of the model

Data dimensionality reduction and feature selection were performed using the least absolute shrinkage and selection operator (LASSO) algorithm [23]. Penalty parameters were adjusted through ten-fold cross-validation and then the robust and non-redundant features were selected. LASSO regression analysis was conducted using R language statistical analysis software. According to the linear equation based on respective coefficients of the selected features, the radiomics model was established, and the radiomics score related to each patient was calculated. Performance of radiomics model was assessed with the area under curve (AUC) values, sensitivity, and specificity.

We also built a DNN model based on the extracted features. Each linear layer in the DNN model is connected by non-linear activation functions to learn complex non-linear relationships. In this paper, we utilized a two-layer neural network with BatchNorm and Dropout modules for better performances. BatchNorm [24] is a normalizing function that reduces the impact of internal covariate shift caused by different scales of training weights. Dropout [25] is a regularizing tool that randomly drops neurons from the neural network during training. Each layer is connected by a rectified linear unit (ReLU) activation function with a dropout rate equal to 0.55. The learning rate is set to 0.001, and all trainable parameters are optimized by the Adam algorithm with batch size as 32 and the number of epoch is set to 50 for training. Training and testing of the DNN were performed on Python (version 3.8).

Model performance was mainly measured by discrimination capability. Receiver operating characteristic (ROC) curve was drawn to quantify the discrimination of the model [26]. In order to determine the clinical value of the model, decision curve analysis (DCA) was performed to reckon the net benefits under different threshold probabilities [27].

Statistical analysis

Statistical analyses were conducted by using SPSS software, and R language statistical analysis software (version 3.4.4). The Kolmogorov-Smirnov test was used to determine the normality of all continuous variables. Univariate analysis (chi-square test for categorical variables and *t*-test for continuous variables) was applied to identify whether PSM controlled the effect of confounding factors. Spearman's rank correlation coefficient was calculated to assess the correlation between the radiomics score and the occurrence of sepsis after surgery. When the p value on both sides was less than 0.05, the difference was considered statistically significant.

Result

Demographic characteristics

Clinical data and medical laboratory results are shown in Tables 1 and 2, respectively. Before PSM, 847 patients met the inclusion criteria, including 549 males (64.8%) and 298 females (35.2%). Total incidence of sepsis after FURL or PCNL was 5.9% (50/847). In univariable analysis, there were significant differences ($p < 0.05$) between the sepsis group and the non-sepsis group in the following indicators: gender, preoperative fever, urine culture, albumin, globulin, albumin globulin ratio, urine white blood cell (qualitative), urine white blood cell (quantitative), urine nitrite. After executing PSM, 132 subjects were successfully matched, including 44 cases in the sepsis group and 88 cases in the non-sepsis group, respectively. No significant differences were found in all parameters between the two groups, as shown in Tables 1 and 2.

Model establishment and estimation

In the radiomics interpretation process, a total of 884 radiomics features of each patient were extracted. The intraobserver and interobserver ICC were beyond 0.7, which demonstrated good test-retest and inter-observer feature extraction reproducibility. Ultimately, all outcomes were derived from the measurement of reader 1. After LASSO analysis, the λ value that provides the best data fit for the model is 0.0202, as shown in Fig. 2. The final model incorporated 26 key features, and the corresponding coefficients of the selected features were used to construct radiomics score calculation formula based on linear equation (Fig. 3). Supplementary Figure 1 shows the distribution of radiomics scores and postoperative infection for each patient with ureteral calculi. According to the correlation analysis, there was a positive correlation between the radiomics score and the occurrence of sepsis after the operation (Spearman's correlation coefficient r = 0.662, p value < 0.001). Namely, a high radiomics score is related to the occurrence of sepsis after the operation. The ROC curve of the LASSO-based model is shown in Fig. 4A, with AUC value of 0.881 (95% CI, 0.813–0.931). The sensitivity is 79.55%, and the specificity is 96.59%. In the independent external validation group of 40 ureteral calculi patients, the trained model achieved an AUC of 0.783 (95% CI, 0.766–0.801), with a sensitivity of 88%, and a specificity of 88% (Fig. 4B). Clinical decision curve in Fig. 5 shows that at most threshold probabilities, the net benefit of using radiomics model to predict postoperative sepsis in patients is higher than that of all patients adopting the treatment plan or no intervention plan.

Figure 6 shows the ROC curve of the DNN model. In general, the trained DNN had an AUC of 0.920 (95% CI, 0.906–0.933) in the internal validation (Fig. 6A). The sensitivity is 85.71%, and the specificity is 94.73%. Due to the limited sample size, overfitting was unavoidable in the previous internal validation. Therefore, we also performed independent external validation in another group of 40 patients with ureteral calculi. In this test set, the DNN model still performed well in the prediction of postoperative sepsis, with an AUC of 0.874 (95% CI, 0.856–0.891), a sensitivity of 77%, and a specificity of 88.67% (Fig. 6B).

Table 1 Comparison of clinical characteristics of sepsis and non-sepsis group before and after propensity score matching

Variable	Unmatched			Matched		
	Sepsis	Non-sepsis	*p* value	Sepsis	Non-sepsis	*p* value
Patients, *n*	50	797		44	88	
Age (yr), mean ± SD	50.96 ± 12.67	48.53 ± 12.73	0.192[a]	50.74 ± 11.6	49.91 ± 12.8	0.712[a]
BMI, mean ± SD, (kg/m^2)	24.05 ± 3.77	24.13 ± 2.98	0.893[a]	24.16 ± 2.8	24 ± 3.8	0.797[a]
Sex, *n* (%)			< 0.001			0.534
Male	17 (34%)	532 (66.8%)		17 (38.6%)	39 (44.3%)	
Female	33 (66%)	265 (33.2%)		27 (61.4%)	49 (55.7%)	
Preoperative stenting, *n* (%)			0.1			0.199
Yes	6 (12%)	33 (4.1%)		6 (13.6%)	6 (6.8%)	
No	44 (88%)	764 (95.9%)		38 (86.4%)	82 (93.2%)	
Surgical method, *n* (%)			0.739			0.538
PCN	23 (46%)	386 (48.4%)		21 (47.7%)	47 (53.4%)	
FURL	27 (54%)	411 (51.6%)		23 (52.3%)	41 (46.6%)	
Stone height, *n* (%)			0.454			0.72
< 20 mm	49 (98%)	764 (95.9%)		43 (97.7%)	85 (96.6%)	
≥ 20 mm	1 (2%)	33 (4.1%)		1 (2.3%)	3 (3.4%)	
Stone laterality, *n* (%)			0.53			0.078
Left	24 (48%)	419 (52.6%)		22 (50%)	58 (65.9%)	
Right	26 (52%)	378 (47.4%)		22 (50%)	30 (34.1%)	
Coronary heart disease, *n* (%)			0.95			1
Yes	1 (2%)	17 (2.1%)		1 (2.3%)	2 (2.3%)	
No	49 (98%)	780 (97.9%)		43 (97.7%)	86 (97.7%)	
Hypertension, *n* (%)			0.808			1
Yes	10 (20%)	171 (21.5%)		9 (20.5%)	18 (20.5%)	
No	40 (80%)	626 (78.5%)		35 (79.5%)	70 (79.5%)	
Diabetes, *n* (%)			0.43			0.836
Yes	5 (10%)	56 (7%)		4 (9.1%)	9 (10.2%)	
No	45 (90%)	741 (93%)		40 (90.9%)	79 (89.8%)	
Pre-operative fever, *n* (%)			< 0.001			0.492
Yes	10 (20%)	34 (4.3%)		8 (18.2%)	12 (13.6%)	
No	40 (80%)	763 (95.7%)		36 (81.8%)	76 (86.4%)	
Hydronephrosis, *n* (%)			0.255			1
Yes	3 (6%)	89 (11.2%)		3 (6.8%)	6 (6.8%)	
No	47 (94%)	708 (88.8%)		41 (93.2%)	82 (93.2%)	

Note: *SD*, standard deviation; *BMI*, body mass index; *PCN*, percutaneous nephrolithotomy; *FURL*, flexible ureteroscopy lithotripsy; [a] *t*-test; others (chi-square test)

Discussion

Clinical variables and laboratory examination cannot provide in vivo stone characteristic information, however, which would be essential for sepsis risk stratification [8]. In this study, we investigated the application of deep learning model based on noninvasive radiomics for preoperative sepsis prediction in ureteral calculi patients. We found that both LASSO and DNN models were shown to be effective in sepsis prediction. Specifically, the DNN model achieved superior prediction capability as opposed to the LASSO-based model (AUC 0.920 vs 0.881). Furthermore, the results of the independent validation set indicated that the DNN model obtained preferable classification efficacy compared with the LASSO-based model (AUC 0.874 vs 0.783).

Intracavitary operation (FURL or PCNL) for ureteral calculi is still plagued by some complications, such as sepsis. In this study, the total incidence of sepsis after FURL or PCNL was 5.9% (50/847), in accordance with previous studies [28, 29]. The univariate clinical analysis before PSM suggested

Table 2 Comparison of medical laboratory results before and after propensity score matching between sepsis group and non-sepsis group

Variable	Unmatched			Matched		Variable
	Sepsis	Non-sepsis	*p* value	Sepsis	Non-sepsis	*p* value
Patients, *n*	50	797		44	88	
Urine culture, *n* (%)			< 0.001			0.901
Positive	24 (48%)	57 (7.2%)		18 (40.9%)	37 (42%)	
Negative	26 (52%)	740 (92.8%)		26 (59.1%)	51 (58%)	
WBC, *n* (%)			0.49			0.069
< 10,000 cells/mL	8 (16%)	733 (92%)		11 (25%)	11 (12.5%)	
≥ 10,000 cells/mL	42 (84%)	64 (8%)		33 (75%)	77 (87.5%)	
Albumin, *n* (%)			< 0.001			0.516
< 35 g/L	14 (28%)	61 (7.7%)		9 (20.5%)	14 (15.9%)	
≥ 35 g/L	36 (72%)	736 (92.3%)		35 (79.5%)	74 (84.1%)	
Globulin, *n* (%)			< 0.001			0.801
< 30 g/L	20 (40%)	532 (66.8%)		18 (40.9%)	34 (38.6%)	
≥ 30 g/L	30 (60%)	265 (33.2%)		26 (59.1%)	54 (61.4%)	
AGR, *n* (%)			< 0.001			0.245
< 1.5	45 (90%)	484 (60.7%)		39 (88.6%)	83 (94.3%)	
≥ 1.5	5 (10%)	313 (39.3%)		5 (11.4%)	5 (5.7%)	
Urine WBC, *n* (%)			< 0.001			0.349
Positive	38 (76%)	378 (47.4%)		11 (25%)	29 (33%)	
Negative	12 (24%)	419 (52.6%)		33 (75%)	59 (67%)	
Urine WBC, *n* (%)			< 0.001			0.196
< 50 cells/mL	13 (26%)	429 (53.8%)		12 (27.3%)	34 (38.6%)	
≥ 50 cells/mL	37 (74%)	368 (46.2%)		32 (72.7%)	54 (61.4%)	
Urine nitrite, *n* (%)			< 0.001			0.53
Positive	9 (18%)	25 (3.1%)		7 (15.9%)	18 (20.5%)	
Negative	41 (82%)	772 (96.9%)		37 (84.1%)	70 (79.5%)	

Note: *AGR*, albumin globulin ratio; *WBC*, white blood cell

that the potential independent variables of postsurgical sepsis included gender, preoperative fever, urine culture, and albumin globulin ratio. Our results are in line with some clinical experience and previous studies [30–32]. A prospective investigation by Blackmur et al [31] showed that positive urine culture result before surgery is a risk factor for postoperative sepsis. Moreover, it is worth noting that females (33/50, 66%) are more susceptible to sepsis than males (17/50,

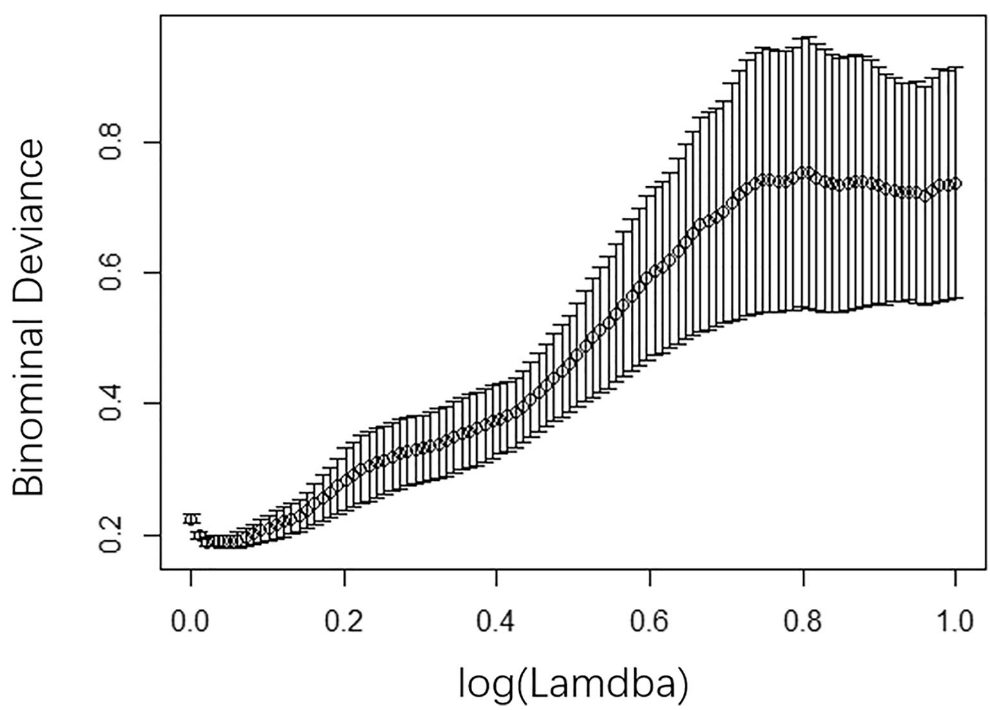

Fig. 2 LASSO regression analysis performs feature selection. Select the parameter (λ) in the LASSO algorithm based on the minimum criterion and 10-fold cross-validation. The *y*-axis represents the binomial deviation. The *x*-axis represents log (λ). The point in the middle of the curve represents the average deviation value of each model corresponding to λ, and the vertical line represents the upper limit and lower limit of the deviation. The final selected λ value is 0.0202, which provides the best data fit for this model

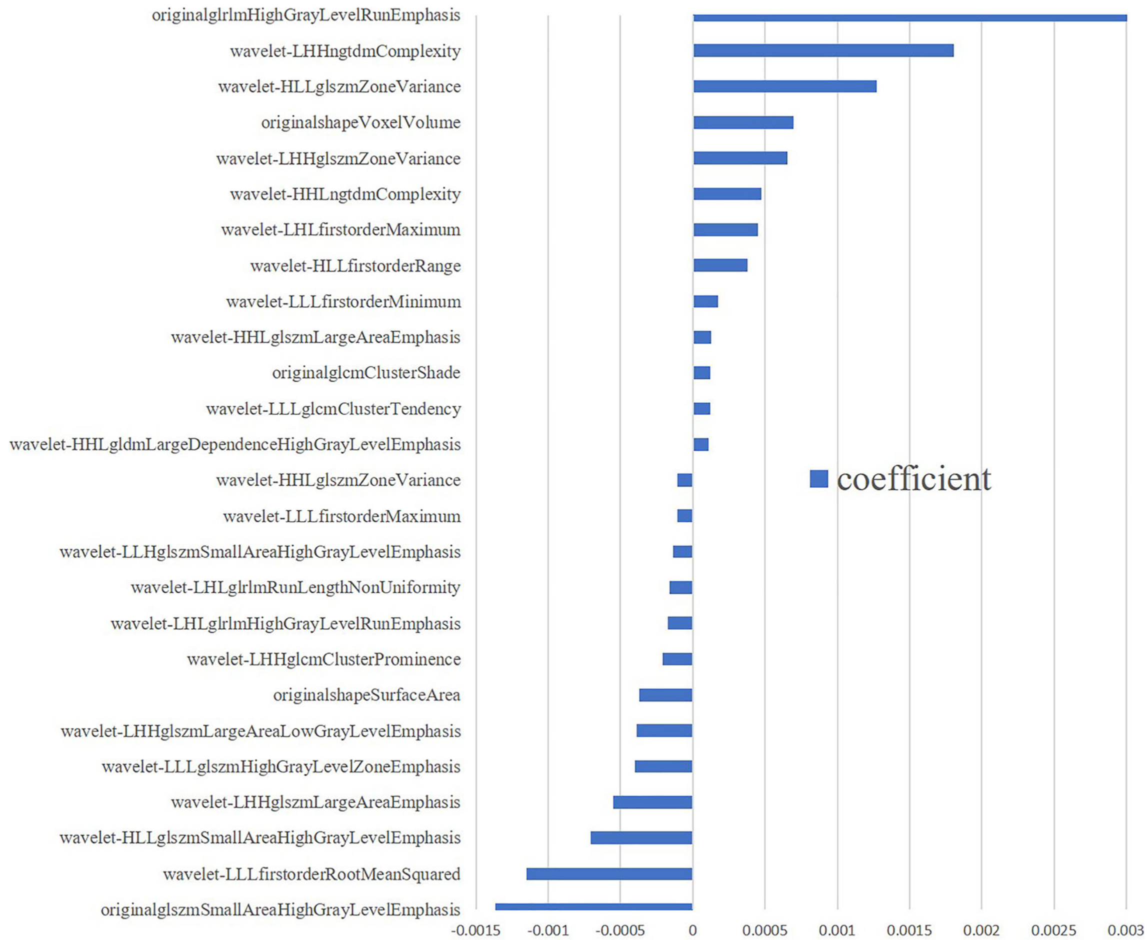

Fig. 3 The most predictive features in the radiomics model and their corresponding coefficients. The larger the absolute value of the coefficient, the more important the feature is

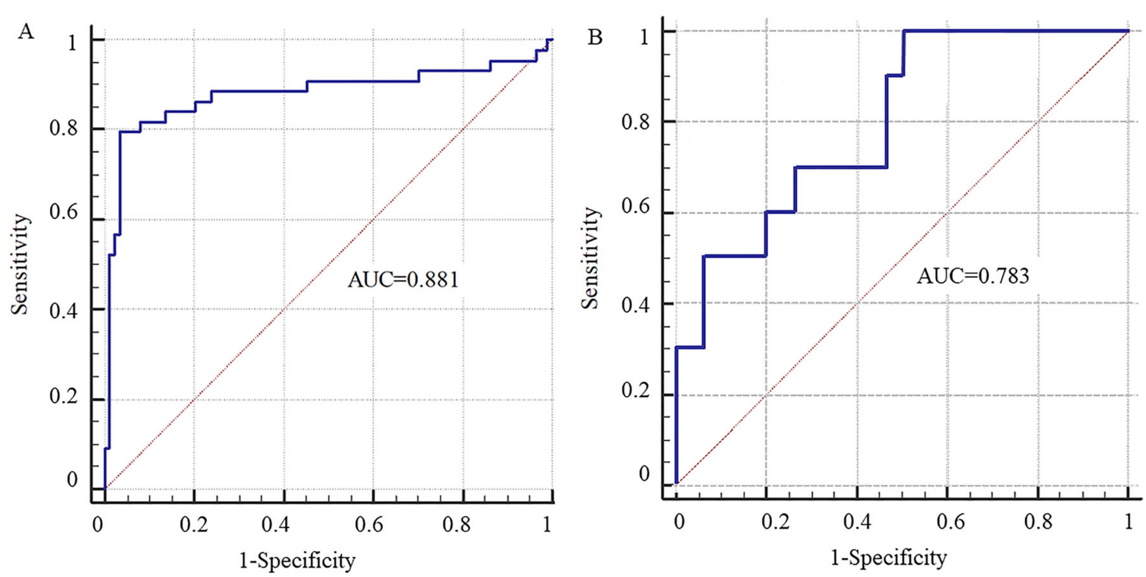

Fig. 4 Receiver operating characteristic (ROC) is used to evaluate the performance of the LASSO model. **A** ROC of the trained LASSO model in the internal validation. The area under the curve (AUC) of the model is 0.881 (0.813–0.931); **B** ROC of the trained LASSO model in the external validation, with the AUC of 0.783 (0.766–0.801)

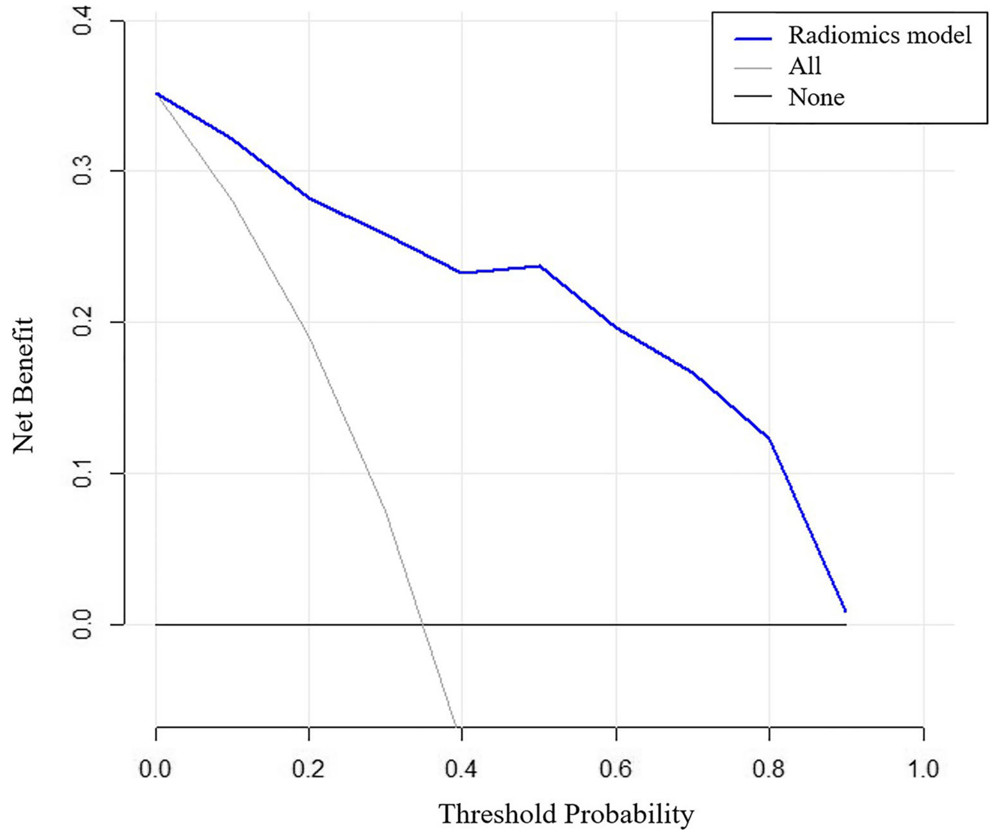

Fig. 5 Decision curve analysis (DCA) was used to evaluate the clinical value of the radiomics model. The *y*-axis measures the net benefit and the *x*-axis represents the threshold probability. The decision curve shows that if the threshold probability was below 0.9, the use of radiomics model to predict the occurrence of postoperative sepsis will benefit more than the "All (All interventions)" or "None (No intervention at all)" strategies

34%) in our research. A possible explanation is that the anatomy of the female urinary tract is relatively short and straight. These high-risk patients should be properly consulted before surgery and should be the focus of postoperative vigilance monitoring.

Pioneering studies have evaluated the performance of deep learning and radiomics in medical diagnosis [33–35], therapy guidance [36], and prognostic assessment [37] in the realm of oncology. Since radiomics can deal with high-throughput features to obtain intra-lesion heterogeneity details [38], it is gradually applied to many other non-tumor diseases, such as urinary calculi [17, 39]. In addition, deep learning has also been employed in the diagnosis and treatment of urinary calculi. For example, Parakh et al [40] attempted to achieve automatic detection of urolithiasis on unenhanced CT images by convolutional neural network (CNN). In our study, radiomics-based image analysis enabled robust differentiation of postoperative sepsis and non-sepsis groups in ureteral calculi

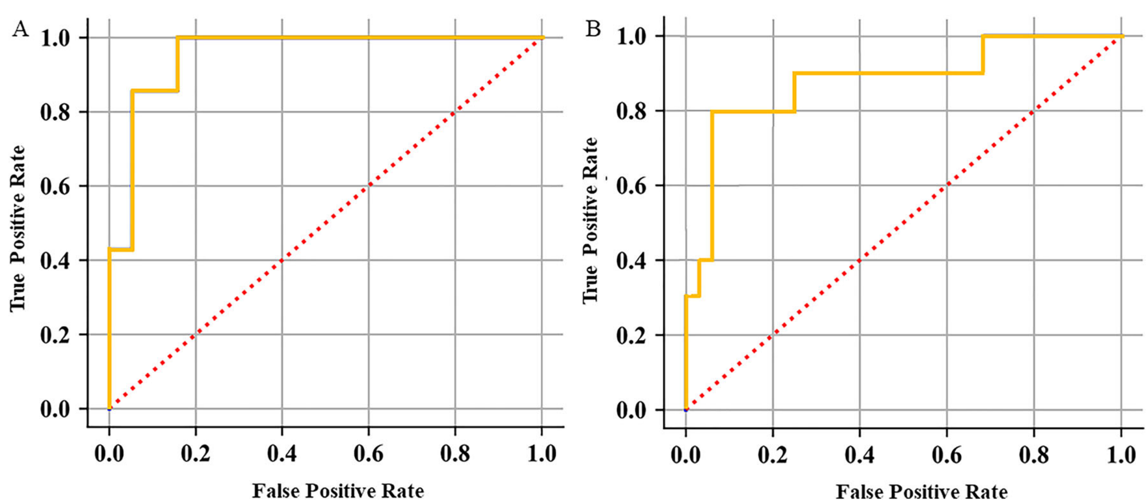

Fig. 6 ROC is used to evaluate the performance of the DNN model. **A** ROC of the trained DNN model in the internal validation. The area under the curve (AUC) of the model is 0.920 (0.906–0.933); **B** ROC of the trained DNN model in the external validation, with the AUC of 0.874 (0.856–0.891)

patients, with an AUC of 0.881 using the LASSO-based model. Furthermore, we found that the predictive model would have better performance when the DNN model was complemented (AUC value increased from 0.881 to 0.920). The adapted deep learning model exhibited even better performance because it captures the high-order interactions among the identified variables. Similarly, Dong et al [36] also approved that DNN performed better than other baseline machine learning (such as support vector machine) methods in predicting the lymph node metastasis of operable cervical cancer. Zhou et al [41] found that deep learning achieved higher diagnostic accuracy than a conventional radiomics method in mass lesions detected on breast MRI. Deep learning applied to biological image analysis is powerful and developing rapidly, but it is often regarded as a black box [42]. Regardless of visualization, almost all of the nomograms were founded on traditional regression methods, and their roles in the era of big data are limited. In the future, the construction of intelligent medical assistance systems, massive medical data will need to be integrated. This may be beyond the scope of traditional machine learning capabilities. However, DNN could automatically assign appropriate weights to each variable based on its contribution without dimensionality reduction, thereby integrating disparate enormous data very efficiently [43]. We believe that the development and application of deep learning-based predictive models combined with clinical data will be the future direction of clinical and scientific research.

There are several limitations in our study. First, in this retrospective study in one center, the research sample is comparatively limited and there was selection bias. Secondly, in order to control confounding factors, clinical risk factors were not included in the prediction model after PSM. Finally, due to technical limitations, only single ureteral stones were included in the study. The future research goal is to prospectively collect cases of urinary calculi based on multi-center study, and combine clinical factors to construct a more robust model.

In summary, this study demonstrates the utility of the radiomics and DNN-based model using pre-therapeutic NCCT images of ureteral calculi patients. The DNN model yields to higher accuracy, and thus has the potential to serve as a guard to promote sepsis risk stratification for ureteral calculi patients.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00330-022-08882-5.

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Declarations

Guarantor The scientific guarantor of this publication is Zhen Li (Department of Radiology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology)

Conflict of interest The authors declare no competing interests.

Statistics and biometry Qingpeng Zhang (School of Data Science, City University of Hong Kong, Kowloon, Hong Kong, China) kindly provided statistical advice for this manuscript.

Informed consent Written informed consent was waived by the Institutional Review Board.

Ethical approval Institutional Review Board approval was obtained.

Methodology
- retrospective
- observational
- performed at one institution

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